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Autoignition behavior of unsaturated hydrocarbons in the low and high temperature regions

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Abstract

In this work, numerical and experimental techniques are used to investigate the effect of the position of the double bond on the ignition properties of pentene and hexene linear isomers. A wide-range kinetic model for the oxidation of C_5–C_6 linear alkenes has been developed. Literature rapid compression machine data were used to validate the model at low temperatures of 660–850 K and new shock tube experiments were performed in order to assess the behavior of the considered alkenes in the high temperature region of 990–1770 K. The experiments and mechanism validations were performed at pressures up to 11 atm to validate the mechanism at conditions found in internal combustion engines.

Some interesting inversions in the relative reactivity of the isomers were detected as the temperature is increased. The model successfully reproduced the measured behavior and allowed insight into the reason for these reactivity changes. The information gathered will be applied to the development of the kinetic mechanisms of larger unsaturated components.
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1. Introduction

Computational fluid dynamics codes with chemical kinetic models are consolidated tools for the investigation of engine performance and the development of more efficient combustion strategies. The increasing power of modern computers makes it increasingly desirable to couple the detailed fluid dynamic modeling with a detailed description of the combustion kinetics [1–3]. For this reason, the development of such models represents a valuable resource for facing the new demands of the energy industry to increase engine efficiency, reduce emissions and maintain engine performance.

Nevertheless, the modeling of real fuel represents a difficult challenge. Real fuels are complex mixture of hundreds of compounds whose exact composition cannot be easily determined. Even knowing the exact formulation of such blends, modeling the huge number of species involved in their oxidation still would be far beyond the present computational capabilities and state of our current knowledge of combustion chemistry. For this reason simpler surrogates are generally adopted.

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For a long time n-heptane and iso-octane have been used as an approximation of real fuels in most of the experimental and numerical investigations of gasoline and diesel fueled engines. The evolution of oil industry is pushing towards new kinds of fuels whose compositions are more and more different from conventional fuels. New kinds of feedstock and the evolution of biofuels are shaping a new generation of fuel requiring the development of sophisticated kinetics models to account for more varied compositions [4–6].

Alkenes are one of the fuel components of interest. Alkenes are formed during the refining of crude oil to gasoline by cracking the heavier fractions. Light alkenes (C4–C5) are generally converted into valuable saturated branched components by alkylation processes. However, both American and European gasolines contain a significant amount of these unsaturated species [7]. Their presence has both advantages and drawbacks: short chain alkenes have a much higher knock resistance than their saturated homologues [8]; on the other hand, at high concentrations they can alter the stability of the fuel and promote the formations of gums, detrimental for engine injectors.

One more reason for interest in this class of compounds is the fact that biodiesels are constituted by most part of unsaturated fatty acid esters. The number of double bonds along the linear carbon chain attached to the ester group affects the cetane number of the fuel and, as of today, very few kinetic models and experimental works have investigated the reactivity of unsaturated bio esters. Additional knowledge about the role of double bonds in fuel chemistry will benefit the understanding of biodiesel unsaturated fatty acid esters.

In this paper, the role of the double bond on ignition properties has been investigated by means of chemical kinetic modeling and experimental techniques. A wide-range kinetic model for the oxidation of C5–C6 linear alkenes has been validated by comparison with literature experimental data. New shock tube experiments were performed in order to assess the behavior of the fuels of interest in the high temperature region at atmospheric and elevated pressures. The effects of the position of the double bond into the alkyl chain are discussed with the help of calculations.

2. Previous works

Contrary to alkanes, the oxidative behavior of alkenes has been relatively little investigated.

The effect of olefin addition on n-heptane cool flames combustion was already noticed in 1971 by Tipper and Titehard [9]. Reviewing work on pentenes, the first systematic study of the low temperature combustion of pentene isomers was carried out by Hughes and Prodhan [10], who investigated the development of cool flames by 1- and 2-pentene in a closed vessel. A study on the low temperature oxidation products of 1 and 2-pentene diluted in a H_2/O_2 mixture was carried out by Baldwin et al. [11] in 1980. Sixteen years later, Prabhu et al. [12] analyzed the oxidation of 1-pentene in a flow reactor at 6 atm from 600 to 800 K at 0.4 equivalence ratio, providing the first speciation data of neat 1-pentene. More recently Minetti et al. compared the ignition behavior of pentane and 1-pentene in a rapid compression machine (RCM) device [13]. These data have been modeled in a following work by Ribaucour et al. [14]. The high temperature ignition of 1-pentene was later discussed in an experimental and modeling work by Touchard et al. [15].

Reviewing work on hexenes, Yahyaoui et al. [16,17] provided speciation data on the oxidative behavior of 1-hexene in a jet stirred reactor and ignition delay times in a shock tube. Kiefer et al. [18] recently measured the decomposition rate of 1-hexene in a shock tube. The first systematic study on the effect of the position of the double bond in hexene isomers was provided by Vanhove et al. [19], who determined the ignition behavior the three hexene isomers in a rapid compression machine at different pressure.

Though the oxidation of alkenes represents an important submechanism in the combustion of higher alkanes, no low temperature mechanisms for this class of compounds were available until recently. The first systematic studies including a low temperature branching mechanism for alkenes were carried out by Mehl et al. [20] in a joint collaboration between Milano and LLNL and by the combustion group in Nancy [21,22].

When comparing the reactivity of the different alkene isomers, the most extensive set of data is the one produced by Vanhove et al. [19]. In this set of experiments is evident how the length of the saturated carbon chain determines the autoignition propensity of alkenes in the low temperature region. Long saturated carbon chains allow the alkyl peroxy isomerization process necessary to sustain a low temperature branching. The set of experiments provided by Vanhove is limited to temperatures lower than 850 K. Mehl et al. [20] computations predicted that at higher temperature the reactivity scale of the three isomers of hexene significantly changes. Unfortunately no experimental information was available to validate this modeling prediction. In this work the autoignition tendency of the pentene and hexene linear isomers was systematically investigated in both the low and the high temperature region, with data up to 1700 K.

3. Experimental

Experiments were carried out in both a high and low pressure shock tube. The high pressure shock tube consists of a stainless steel tube,

8.76 m in length, with an internal diameter of 6.3 cm. A double-diaphragm section divides the shock tube into a 3 m long driver section and a 5.73 m test section. Polyester films were used as diaphragms in all experiments, and shock waves were created by self-rupture of the diaphragms. The driver gas used was helium (99.99% pure; BOC). The operational pressure limit of the shock tube is approximately 60 bar. The diagnostic system involves four pressure transducers. The velocity of the incident shock wave is measured at three locations and is then extrapolated to the end-plate. The pressure at the end wall was monitored by a pressure transducer (PCB, 113A24). The ignition delay time was defined as the time between the shockwave arriving at the endwall and the maximum rate of rise of the pressure signal due to ignition. Four examples of typical pressure profiles are shown in Supplementary data. Any pressure profiles exhibiting significant pre-ignition pressure rise were excluded from the study. Prior to the pentene and hexene isomer experiments, the shock tube facility was validated by reproducing experimental results of n-heptane and iso-octane ignition at 10 and 20 bar, performed by the Hanson [23] and Oehlschlaeger [24] groups. These results are also included in Supplementary data, with our data showing good agreement with the previous studies. Some additional details on this facility can be found in Black et al. [25].

The low pressure shock tube has a stainless steel tube with a test section measuring 6.22 m in length, with an internal diameter of 10.24 cm, and a barrel-shaped driver section measuring 53 cm in length. The two sections are separated by a polycarbonate diaphragm, which bursts when forced into contact with a cross shaped cutter due to the pressure differential between the high pressure driver section and the low pressure test section. The driver gas used was helium (99.99% pure; BOC). The diagnostic system involves four pressure transducers. The velocity of the incident shock wave is measured at three locations and is then extrapolated to the end-plate. The pressure at the end wall was monitored by a pressure transducer (Kistler, 603B). Light emission at 431 nm was detected through a fused silica windows embedded in the end-plate using a photodetector (Thorlabs Inc. PDA 55-EC) and a narrow band-pass filter centered at 430 nm with a full-width half-maximum of 10 nm. The ignition delay time was defined as the time between the shockwave arrival at the endwall and the maximum rate of rise of the emission signal.

The 1-pentene (99.5% pure) was supplied by TCI, and trans-2-pentene (99% pure), 1-hexene (99% pure), trans-2-hexene (97% pure) and trans-3-hexene (99% pure) were supplied by Sigma–Aldrich and de-gassed through a series of freeze–thaw–pump cycles, after which no more gas was observed to escape on thawing the solid. The oxygen (99.5% and 99.999% pure), argon (99.999% pure) and nitrogen were supplied by BOC. Mixtures were prepared using the method of partial pressures and their compositions are provided in Supplementary data.

Pentene isomer experiments have been performed at equivalence ratios of 0.5, 1.0 and 2.0, over a reflected shock pressure range of 0.93–1.16 atm and 7.8–10.8 atm and over a temperature range of 993–1770 K. Hexene isomer experiments have been carried out at an equivalence ratio of 1.0, pressures of 8.5–12.1 atm and temperatures from 990 to 1460 K. The estimated uncertainties of all experiments, considering pressure deviations and definition of ignition delay times from pressure and light emission signals, were within 15%.

The incident shock velocity at the endwall was used to calculate the temperature and pressure of the mixtures behind the reflected shock wave using the equilibrium program Gaseq [26]. The thermodynamic data needed for the program of the pentene and hexene isomers were calculated using the THERM program of Ritter and Bozzeli [27], based on group additivity methods developed by Benson [28]. These data are in agreement with estimates in the NIST WebBook [29].

4. Modeling approach

A previous modeling activity carried out in collaboration with the combustion group in Milano already investigated the oxidation of hexenes in a wide range of conditions [20]. The kinetic model developed at that time was based on the Milano's base chemistry and their lumped species approach [30]. A limited amount of experimental data was available at high temperature for all the three isomers so the direct comparison was mainly focused in the low temperature region where more data was available. A more careful analysis of the high temperature region is here presented in the framework of the LLNL fully detailed approach and base chemistry.

The model here presented is a development of the kinetic mechanism compiled collaboratively at LLNL and the National University of Ireland, Galway. The mechanism is based on a C_1–C_4 detailed core detailed mechanism which has been recently improved by Curran and coworkers [23,24]. On the top of this model, different modules have been built accounting for different cases of compounds: linear, branched and aromatic hydrocarbons. The alkene mechanism here discussed is build upon the same reaction classes presented in Mehl et al. [20]. However, the reaction rates for abstraction reactions, radical decompositions, internal isomerizations, and initiation reactions are instead based on previous works by Curran and coworkers [31–33] and Mehl et al. [34]. In the following paragraphs the main oxidation

pathways of alkenes are presented, further details on the reaction rates adopted for each step can be found in the papers previously referred. The complete list of reactions is provided as Supplementary data.

4.1. Low temperature reactivity

The low temperature oxidation of the linear alkenes is dominated by the chemistry of the allyl radical. Firstly, the weak allylic C–H bond makes the abstraction reaction on this site extremely favored. Depending on the position of the double bond, and the length of the residual saturated chain, low temperature reactions may occur. The availability of a saturated chain with a least three carbons makes possible facile isomerization of the hydroperoxyl radicals (ROO) to the peroxyalkyl (QOOH) radicals though a six-membered transition state. This reaction opens the way to the formation of ketohydroperoxides and low temperature branching production of OH radicals. For this reason 1-alkenes have the highest reactivity among the large *n*-alkene isomers.

This behavior has been experimentally detected in several occasions, including by Hughes and Prodhan [10] in the early '70s and by Vanhove et al. [19] in more recent times. The low temperature branching pathway typical of the alkanes competes with a unique characteristic of alkene chemistry: the OH addition on the double bond. This addition rate has a negligible activation energy and is extremely favored at low temperature. The resulting adduct (HOR.H) can undergo O_2 addition and subsequently, decompose via Waddington mechanism to aldehydes and a OH radical. This path is chain propagating, producing one OH radical compared to the previously described chain branching pathway which produces two OH radicals.

The OH radicals produced in the above reaction paths react with alkenes to produce allyl radicals that are resonantly stabilized and relatively unreactive compared to alkyl radicals. Due to their unreactivity, they accumulate in large concentrations. At temperatures below about 1000 K, they are mainly consumed by reaction with HO_2 radicals. This reaction forms reactive OH radicals and alkenoxy radicals. The later readily undergoes beta decompositions to smaller oxygenated structures.

4.2. High temperature reactivity

When the temperature is higher than 900 K, the low temperature pathways become less effective. The addition of O_2 on alkenyl radicals is no longer favored and the β-decomposition of alkenyl radicals takes over. The presence of double bonds inhibits some cleavages and favors the formation of resonantly stabilized fragments. When the temperature further increases, unimolecular alkene decomposition (initiation) becomes competitive with H-atom abstractions and the composition of the radical pool is totally determined by the relative strength of the C–C bonds along the fuel molecule. Figure 1 summarizes the different mechanisms that take place across the increasing temperature ranges.

The transition from a low to high temperature oxidation mechanisms is responsible for the unexpected inversions in reactivity of the isomers with increasing temperature observed in the experimental and modeling results, as will be discussed in the following section.

5. Results

The low temperature data presented here were collected by Vanhove et al. [19] for the hexane iso-

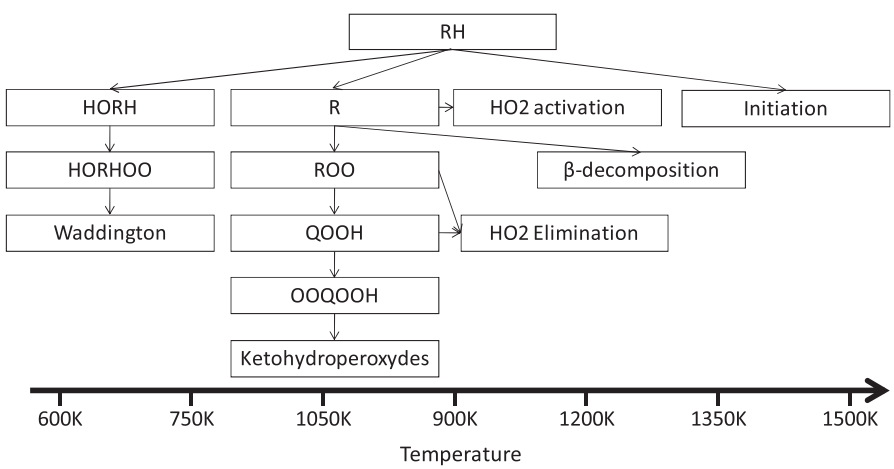

Fig. 1. Reaction pathways determining the radical pool at different temperatures.

mers and Minetti et al. [13] for 1-pentene in the Lille rapid compression machine facility. All the high temperature ignition delay times here presented were recently collected in Galway using the two shock tube described in the experimental section and have not been reported previously. Other ignition experiments for 1-pentene and 1-hexene were previously carried out, respectively, by Touchard et al. [15] and Yahyaoui et al. [16,17]. The shock tube data measured in Galway correlate well with the ones presented in these earlier works, with the exception of Touchard et al. [15], where the activation energy shown by the earlier experiments is higher. Additional comparisons with those data and with selectivity data of hexenes measured by Vanhove et al. [19] are provided as Supplementary data.

All the simulations were performed using the commercial CHEMKIN PRO code and the Milano's DSMOKE code [35]. The shock tube data were simulated assuming that the conditions behind the shock wave were adiabatic, constant volume conditions with no inhomogeneities. The inflection point in the computed temperature profile was used to indicate the ignition delay time.

Since the low temperature autoignition of alkenes typically require long delays, rapid compression machine (RCM) simulations have to account for the heat loss affecting the experimental measurements. Non reactive pressure traces were used to calibrate the heat loss coefficient. Since all the RCM data were collected in the same device at similar conditions, the same parameters have been adopted for all the simulations. This approximation affected mostly the longest ignition delay times, while the effect of heat loss on most of other conditions of interest can be reasonably assumed to be negligible.

5.1. Hexene isomers

Figure 2 shows the comparisons between calculated and measured ignition delay times at about 10 atm over a wide range of temperature covered by the experiments in the RCM and shock tube. The oxidizer is simulated air and the equivalence ratio equal to 1.

The model correctly reproduces the ignition delay times over the whole range of conditions for all the three isomers. The position of the double bond determines the effectiveness of the low temperature branching mechanism. According to the experimental data, 1 and 2-hexenes show some cool flame formation resulting in a negative temperature coefficient behavior, 3-hexene does not have this possibility. The model successfully reproduces these features. A reaction pathway analysis performed by the authors showed that in the temperature range comprised between 650 K and 800 K the main consumption pathways of 1 and 2-hexene are triggered by the low temperature branching. Both these two isomers can form hex-1-en-3-yl radicals by the abstraction of secondary or primary allylic hydrogens. Hex-1-en-3-yl can undergo O_2 addition and start the pathway leading to ketohydroperoxides.

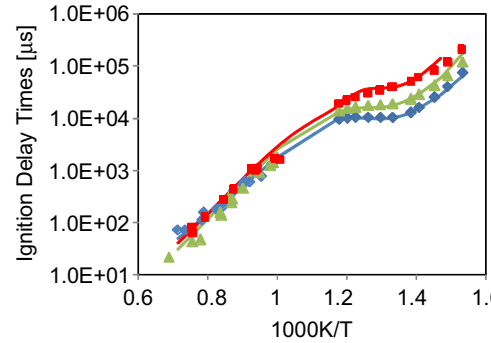

Fig. 2. Ignition delay times of hexene isomers in synthetic air at high pressure: 10 atm, stoichiometric conditions. The curves are results of the chemical kinetic model. The experimental data are given by 1-hexene (diamonds), 2-hexene (triangles), 3-hexene (squares). Low temperature data are from [19].

In the case of 3-hexene the attack on to the double bond and the following decomposition to small oxygenates are the main oxidation routes rather than typical low temperature reaction pathways. The most energetically favored abstractions are on the secondary allylic hydrogens. The resulting hex-2-en-4-yl radicals hardly undergo the same pathway previously described because of the lack of secondary hydrogens available for the six member isomerization of the ROO structures. Direct elimination of HO_2 and diolefin formation is then preferred. Further details on this mechanism can be found in Mehl et al. [20].

When the temperature increases, decomposition reactions and HO_2 chemistry take over the degenerative branching path and the relative reactivity of the three fuels changes.

1-Hexene, the fastest reacting isomer of the three isomers at low temperature conditions, appears to be the slowest reacting isomer for temperatures above 1400 K while 2-hexene becomes the fastest reacting. Figure 3 shows in better detail the high temperature region here studied. The reversed reactivity shown by the experiments can be explained by the population of the radical pool generated by the different isomers. When temperature reaches 1000 K, the decomposition of the resonantly stabilized hexenyl radicals become the main consumption pathway of the fuel. The hex-1-en-3-yl formed by 1 and 2-hexene forms butadiene and highly reactive ethyl radicals. Hex-2-en-4-yl radical is instead the precursor of the slower methyl radical. When the temperature rises to 1400 K, initiation reaction (hexene decomposition) dominates and the position of the double

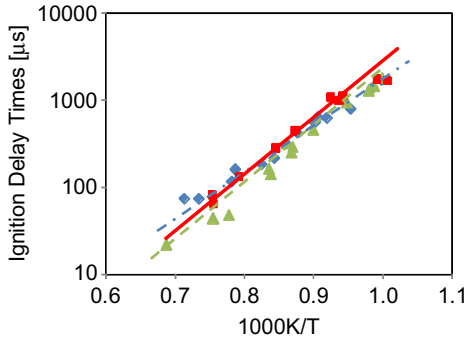

Fig. 3. Ignition delay times of hexene isomers in air at high temperature: 10 atm, stoichiometric conditions. 1-Hexene (Exp: diamonds, Calc: dot-dashed line), 2-hexene (Exp: triangles, Calc: dashed line), 3-hexene (Exp: squares, Calc: solid line).

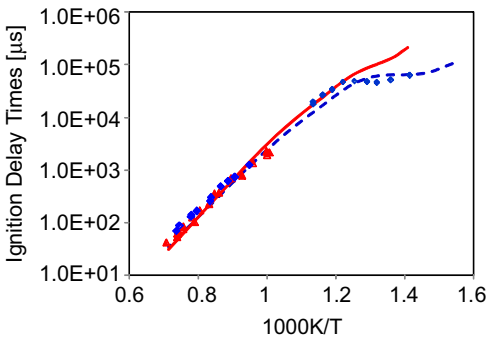

Fig. 4. Ignition delay times of pentene isomers in synthetic air at high pressure: 7–10 atm, stoichiometric conditions. 1-Pentene (Exp: diamonds, Calc: dashed line) and 2-pentene (Exp: triangles, Calc: solid line). Low temperature data are from [13].

bond dictates the composition of the fragments depending on the weakest C–C bond. 1-hexene forms the slowly reacting allyl and 1-propyl radicals (the last of the two ultimately decomposes to ethylene and methyl), while 2-hexene decomposes to ethyl and but-1-en-3-yl radicals: both of the two fragments stabilize by eliminating H radicals that speed the reactivity. 3-Hexene decomposition leads to the formation of butadiene and two methyl radicals, which confer to this fuel a reactivity slightly superior to the one of 1-hexene.

5.2. Pentene isomers

Figure 4 shows the calculated ignition delay times of the two pentene isomers here considered: 1 and 2-pentene. The RCM data [13] covering the 650–850 K temperature range were collected between 7 and 9 bar, while the shock tube data at about 10 bar. Unfortunately no RCM data were available for 2-pentene. Modeling results suggest that differently from 1-pentene, which demonstrated a slight NTC behavior, 2-pentene shows extremely reduced low temperature reactivity. The slower reactivity of 2-pentene is consistent with the findings of Hughes and Prodhan [10]. A reaction pathway analysis showed that the only low temperature reactivity shown by 2-pentene comes from the pent-1-en-3-yl radical formed by the abstraction of the primary allyl hydrogens. Most of the consumption of the 2-pentene comes from the radical addition to the double bond, similar to the case of 3-hexene. When the temperature increases the reactivity of the two isomers tends to become the same.

In order to confirm the validity of the model in the high temperature region some validation at lower pressure was also performed. Figure 5 shows the calculated ignition delay times predicted by the model in comparison with experiments carried out in the low pressure shock tube previously described. All the data were collected at atmospheric pressure and fixed fuel concentration.

The model reproduces fairly well the experimental ignition delay times in all the conditions. One discrepancy at the highest equivalence ratio is that the model tends to slightly over predict

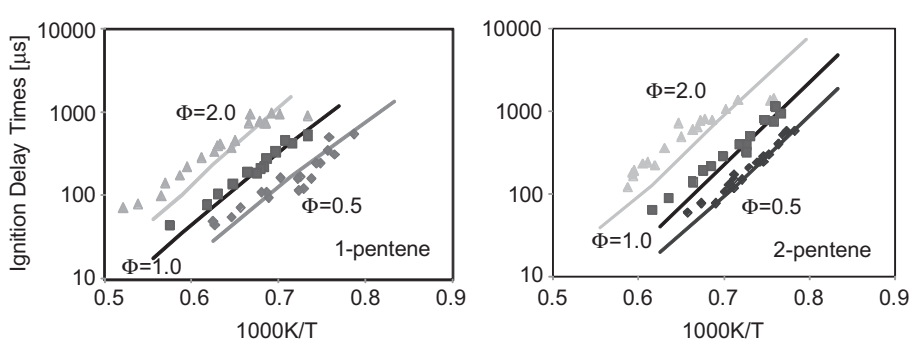

Fig. 5. Ignition delay times of 1 and 2-pentenes in the low pressure shock tube at different equivalence ratios and atmospheric pressure, 1% fuel in argon: Phi = 2 (triangles), 1 (squares), 0.5 (diamonds).

the activation energy of the ignition delay times. The results are still in line with the experiments and support the validity of the results obtained at higher pressure.

The reason why in the case of pentenes is not possible to observe a clear reactivity inversion as for the hexene isomers, but rather a collapse of the two profiles, is that at high temperature the decomposition of both the two isomers leads to the formation of H radicals. At high temperature 1-pentene readily decomposes to an allyl and an ethyl radical, which forms ethylene and H. 2-Pentene leads to methyl and but-1-en-3-yl. The only way this radical can stabilize is through the elimination of an H radical as well. As a result, the high temperature reactivity of the two pentenes is nearly the same.

6. Conclusions

In this paper the autoignition behavior of the pentene and hexene isomers was systematically analyzed by means of experiments and kinetic simulations over a wide range of temperatures and for pressures of 1 and 7–10 atm. The reactivity of this class of compounds was experimentally measured in a temperature region scarcely investigated in previous works. Some peculiar inversions in the relative reactivity of the isomers over the temperature range considered were detected. The model successfully reproduced the measured behavior and provided an interpretative tool for a better understanding of the chemistry of alkenes. The position of the double bond proved to be the determining factor the reactivity of the different isomers not just in the low temperature region, but also at high temperature. The obtained results can be summarized as follows:

(i) At low temperature, the reactivity of large alkenes depends on the length of the saturated portions aside the double bond: longer alkyl chains determine shorter ignition delay times, as already discussed in previous works [19–21].
(ii) At about 1000 K, the reactivity is determined by the radicals formed by the β-decompositions of the resonantly stabilized radicals, the most abundant in the system. Fragments leading to the formation of H radicals, such as ethyl radicals, promote the high temperature radical branching.
(iii) At temperatures above 1400 K, the oxidation process is driven by initiation reactions. The presence of the double bond makes the C–C cleavages forming allyl radicals the most energetically favored. Once again the population of the resulting radical pool determines the reactivity of the fuel. Fragments decomposing to alkenes and H radicals tend to be faster than methyl radicals.

The pieces of information collected in this work provided a validation of the reaction rate rules adopted in the LLNL kinetic model. The same rate constants will be applied for the development of the mechanisms for larger alkenes and unsaturated fatty acid esters contained in biofuels.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.proci.2010.05.040.

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